Static Stress Analysis

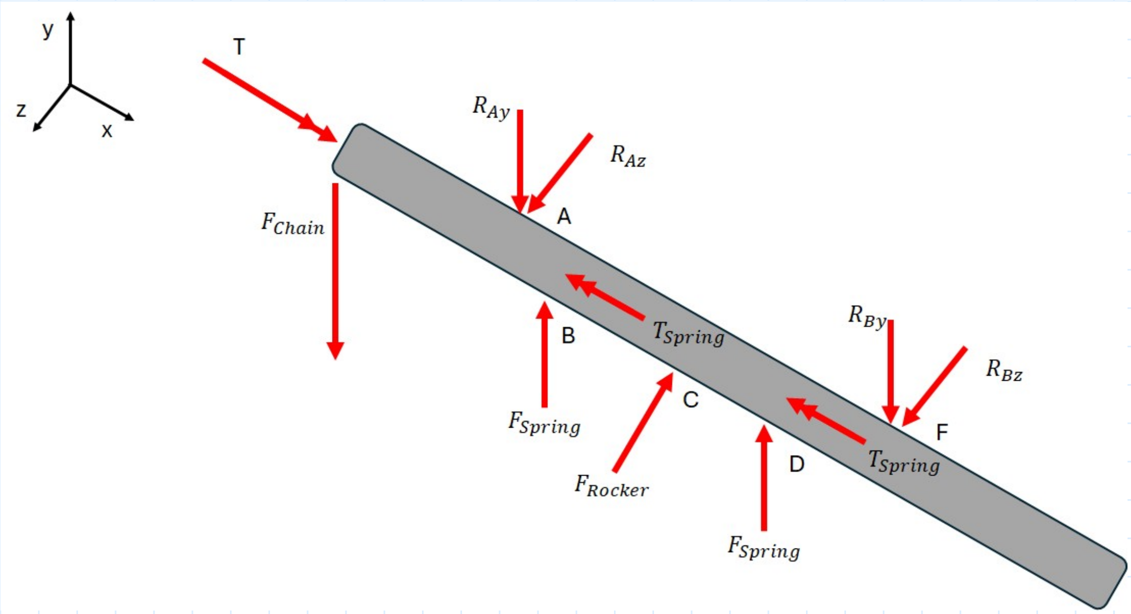
Camshaft

Material Properties

8620 Steel Per ASTM A29

$\sigma_{ult} := 530\ MPa$	Ultimate Tensile Strength
$\sigma_{yield} := 385\ MPa$	Yield Tensile Strength
$E := 200\ GPa$	Modulus of Elasticity
$G := GPa$	Modulus of Rigidity
$\nu := 0.30$	Poisson's Ratio

Free Body Diagram



Dimensions

$L := 118\ mm$

$d := 17\ mm$

$A := 29\ mm$

$B := 41.66\ mm$

$$I := \pi \cdot \frac{d^4}{64} = \left(4.1 \cdot 10^3\right)\ mm^4$$

$C := 59.9\ mm$

$D := 77.09\ mm$

$F := 90\ mm$

$$J := 2\ I = \left(8.2 \cdot 10^3\right)\ mm^4$$

$F_{Open} := 668\ N$

$F_{Closed} := 223\ N$

$F_{Spring} := F_{Open} = 668\ N$ — Force of Spring at Max Open Length

$Rocker_{MA} := \dfrac{40\ mm}{25\ mm} = 1.6$ — Mechanical Advantage from Rocker Arm

$F_{Rocker} := \dfrac{2 \cdot F_{Spring}}{Rocker_{MA}} = 835\ N$ — Force Applied on Camshaft From Rocker Arm

$r_{CamLobe} := 15\ mm$

$T_{Cam} := \left(2 \cdot F_{Spring}\right) \cdot r_{CamLobe} = 20.04\ N \cdot m$

$r_{Gear} := \dfrac{70.5\ mm}{2} = 35.25\ mm$

$F_{Chain} := -\dfrac{T_{Cam}}{r_{Gear}} = -568.511\ N$

Solving for Reaction Forces at Bearing Locations in Y-Direction

Guess Values

$R_{Ay} := 1\ N$

$R_{By} := 1\ N$

Constraints

$2 \cdot F_{Spring} + F_{Chain} - R_{Ay} - R_{By} = 0$

$-\left(R_{Ay} \cdot A\right) + \left(F_{Spring} \cdot B\right) + \left(F_{Spring} \cdot D\right) - \left(R_{By} \cdot F\right) = 0$

Solver

$$\begin{bmatrix} R_{Ay} \\ R_{By} \end{bmatrix} := \textbf{find}\left(R_{Ay}, R_{By}\right)$$

$R_{Ay} = -168.048\ N$ $\quad R_{By} = 935.538\ N$

Solving for Reactions Forces at Bearing Locations in Z-Direction

Guess Values

$$R_{Az} := 1\ N$$

$$R_{Bz} := 1\ N$$

Constraints

$$R_{Az} + R_{Bz} - F_{Rocker} = 0$$

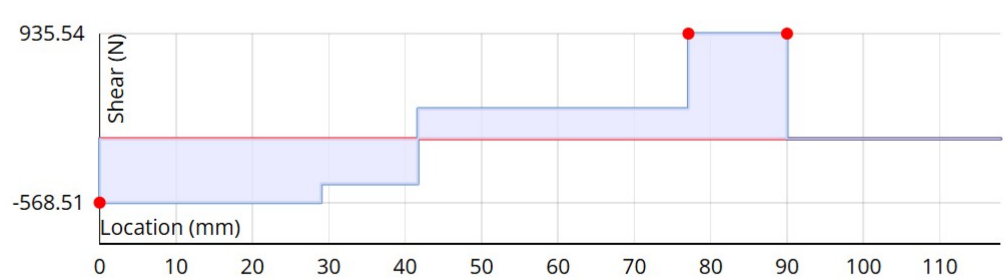
$$\left(F_{Rocker} \cdot (C - A)\right) - \left(R_{Bz} \cdot (F - A)\right) = 0$$

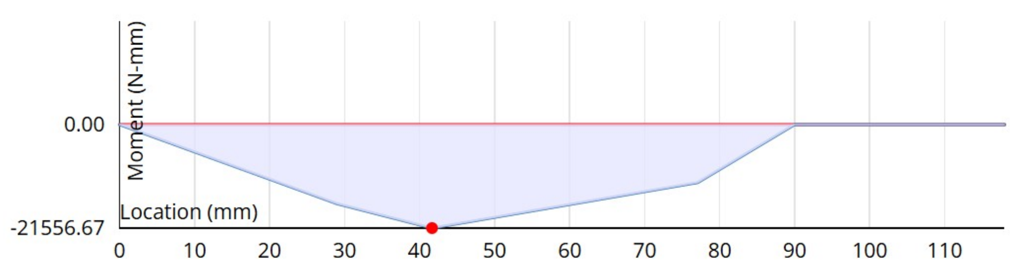
Solver

$$\begin{bmatrix} R_{Az} \\ R_{Bz} \end{bmatrix} := \text{find}\left(R_{Az}, R_{Bz}\right)$$

$$R_{Az} = 412.025\ N \qquad R_{Bz} = 422.975\ N$$

Shear and Moment Diagrams XY-Plane



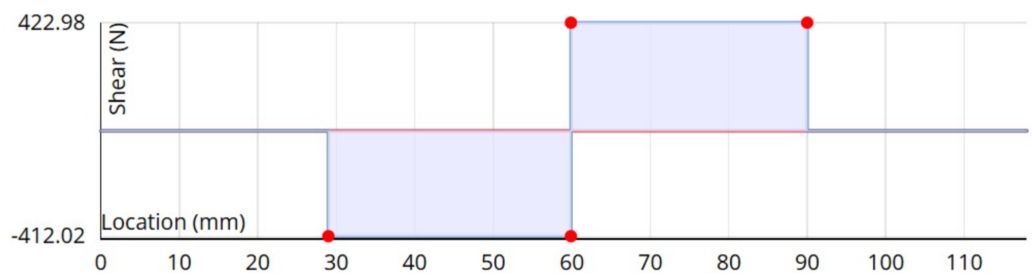


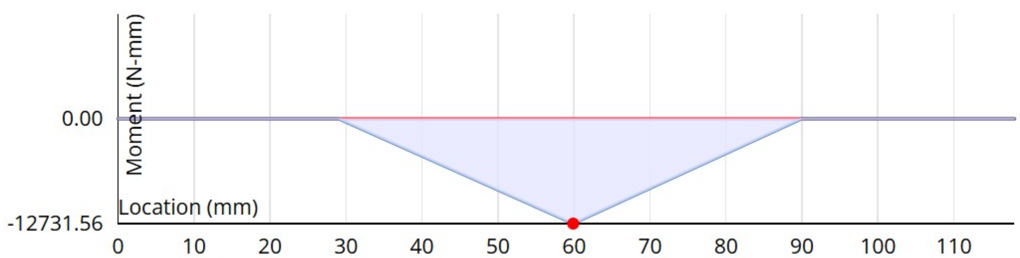
$M_{Max\_z} := 21556.67\ N \cdot mm$

Max Moment in z occurs at B

$M_{z\_C} := M_{Max\_z} - (F_{Chain} - R_{Ay} + F_{Spring}) \cdot (C - B) = 16.677\ N \cdot m$

Shear and Moment Diagrams XZ-Plane





$M_{Max\_y} := 12731.56\ N \cdot mm$

Max Moment in y occurs at C

$M_{y\_B} := R_{Az} \cdot (B - A) = (5.216 \cdot 10^{3})\ N \cdot mm$

Determining Critical Cross-Section

$$M_{R\_B} := \sqrt{(M_{Max\_z})^2 + (M_{y\_B})^2} = (2.218 \cdot 10^{4})\ N \cdot mm$$

$$M_{R\_C} := \sqrt{(M_{Max\_y})^2 + (M_{z\_C})^2} = (2.098 \cdot 10^{4})\ N \cdot mm$$

$$Moment_{Max} := \max(M_{R\_B}, M_{R\_C}) = (2.218 \cdot 10^{4})\ N \cdot mm$$

Critical Cross-Section is located at Point B

Determining Stresses

$$\sigma_{bend} := \frac{Moment_{Max} \cdot 0.5 \cdot d}{I} = 45.982 \ MPa$$

$$\tau := \frac{T_{Cam} \cdot 0.5 \cdot d}{J} = 20.774 \ MPa$$

$$\tau_{max} := \frac{16}{\pi \cdot d^3} \cdot \sqrt{\left(Moment_{Max}\right)^2 + \left(T_{Cam}\right)^2} = 30.986 \ MPa$$

$$\sigma_{vm} := \sqrt{\sigma_{bend}^{\ 2} + 3 \cdot (\tau)^2} = 58.387 \ MPa$$

Safety Factor

$$FS_{yield} := \frac{\sigma_{yield}}{\sigma_{vm}} = 6.594$$

$$FS_{ult} := \frac{\sigma_{ult}}{\sigma_{vm}} = 9.077$$

Bearing Safety Factor (Assuming an axial load equal to the radial load)

Bearing

PN: SKF 7203 BEP

$C := 10.4\ kN$ Basic Dynamic Load Rating

$C_0 := 5.5\ kN$ Basic Static Load Rating

$P_u := 0.236\ kN$ Fatigue Load Limit

$RPM_{max} := 20000\ rpm$ Max RPM

$X_0 := 0.35$ Radial Load Factor

$Y_0 := 0.26$ Axial Load Factor

$e := 1.1$ Limiting Factor

$p := 3$ Exponent of the Life Equation

$a_1 := 0.64$ Life Adjustment Factor at 95% Reliability

$a_{SKF} := 25$ Life Modification Factor (Best Estimate)

$n := 12500\ rpm$

Bearings are located at locations A and F, the loads on the bearings are captured in the Reaction forces above.

$$F_{bearing\_A} := \sqrt{{R_{Ay}}^2 + {R_{Az}}^2} = 444.977\ N$$

$$F_{bearing\_F} := \sqrt{{R_{By}}^2 + {R_{Bz}}^2} = \left(1.027 \cdot 10^3\right)\ N$$

$F_{bearing\_max} := \max\left(F_{bearing\_A}, F_{bearing\_F}\right) = \left(1.027 \cdot 10^3\right)\ N$ Max Radial Load

$F_{axial} := F_{bearing\_max} = \left(1.027 \cdot 10^3\right)\ N$ Axial Load

$P_0 := X_0 \cdot F_{bearing\_max} + Y_0 \cdot F_{bearing\_max} = 626.295\ N$ Equivalent Static Bearing Load

$S_0 := \dfrac{C_0}{P_0} = 8.782$ Bearing Static Safety Factor

Fatigue Analysis

Camshaft

Values from Static Analysis

$\sigma_{bend} = 45.982\ MPa$

$I = \left(4.1 \cdot 10^3\right)\ mm^4$

$\sigma_{vm} = 58.387\ MPa$

$J = \left(8.2 \cdot 10^3\right)\ mm^4$

$\tau = 20.774\ MPa$

Assumptions

-Shaft Continuously Rotating and Always Same Direction

-Fully Reversing Bending (More Conservative)

$\sigma'_a := \sqrt{\sigma_{bend}^{\ 2} + 0} = 45.982\ MPa$ Alternating Stress

$\sigma'_m := \sqrt{0 + 3\ \tau^2} = 35.982\ MPa$ Mean Stress

$S_e := 0.4 \cdot \sigma_{ult} = 212\ MPa$ Endurance Limit

Fatigue Factor of Safety

(Shigley's Table 6-8)

$$n_f := 0.5 \cdot \left(\frac{\sigma_{ult}}{\sigma'_m}\right)^2 \cdot \left(\frac{\sigma'_a}{S_e}\right) \cdot \left(-1 + \sqrt{1 + \left(\frac{2 \cdot \sigma'_m \cdot S_e}{\sigma_{ult} \cdot \sigma'_a}\right)^2}\right) = 4.23$$

Results

Although analysis was performed statically, the large safety factor margins of both fatigue and yield stress for the camshaft provide confidence in the addition of dynamic loading caused by high RPM engine operation.

Fatigue Analysis

Valve Spring

Material Properties

Chromium-Silicon Alloy Steel Wire Per ASTM A401

$m := 0.108$ $\qquad A := 1974\ MPa \cdot mm^m$ $\qquad d := 3.0\ mm$

$$\sigma_{ult} := \frac{A}{d^m} = \left(1.753 \cdot 10^3\right)\ MPa$$ (Shigley's Table 10-4)

$$\tau_{max} := \frac{2}{3} \cdot \sigma_{ult} = \left(1.169 \cdot 10^3\right)\ MPa$$

$$D := 19\ mm$$

$$C := \frac{D}{d} = 6.333$$

$$K_B := \frac{4 \cdot C + 2}{4 \cdot C - 3} = 1.224$$

$$\tau_{upper} := K_B \cdot \frac{8 \cdot F_{Open} \cdot D}{\pi \cdot d^3} = \left(1.465 \cdot 10^3\right)\ MPa$$

$$\tau_{lower} := K_B \cdot \frac{8 \cdot F_{Closed} \cdot D}{\pi \cdot d^3} = 489.073\ MPa$$

$$\tau_m := \frac{\tau_{upper} + \tau_{lower}}{2} = 977.05\ MPa$$ Mean stress

$$\tau_a := \frac{\tau_{upper} - \tau_{lower}}{2} = 487.977\ MPa$$ Average Stress

$$S_u := \tau_{max} = \left(1.169 \cdot 10^3\right)\ MPa$$

$$S_e := 0.4 \cdot S_u = 467.506\ MPa$$

$$n_f := \frac{1}{\dfrac{\tau_a}{S_e} + \dfrac{\tau_m}{S_u}} = 0.532$$ Fatigue Factor of Safety

Results

Since the fatigue factor of safety is less than 1, we can expect the spring to fail with the given parameters. Since the torsional stress in wire coil springs is highly dependent on the wire diameter, we will solve for a needed wire diameter to satisfy a fatigue safety factor of 2, however, I will not take into account the calculation of the ultimate strength due to its dependence on d, which will lower the safety factor from the requirement. Iterations will be performed to achieve FOS=2.

$$d := \sqrt[3]{\frac{4 \cdot K_B \cdot D \cdot \left(\frac{F_{Open} - F_{Closed}}{S_e} + \frac{F_{Open} + F_{Closed}}{S_u} \right)}{0.5 \cdot \pi}}$$

$d = 4.665\ mm$ Calculated wire diameter

$d := 5\ mm$ Iterative wire diameter

Verification

$$\sigma_{ult} := \frac{A}{d^m} = \left(1.659 \cdot 10^3\right)\ MPa$$ (Shigley's Table 10-4)

$$\tau_{max} := \frac{2}{3} \cdot \sigma_{ult} = \left(1.106 \cdot 10^3\right)\ MPa$$

$D := 19\ mm$

$$C := \frac{D}{d} = 3.8$$

$$K_B := \frac{4 \cdot C + 2}{4 \cdot C - 3} = 1.41$$

$$\tau_{upper} := K_B \cdot \frac{8 \cdot F_{Open} \cdot D}{\pi \cdot d^3} = 364.526\ MPa$$

$$\tau_{lower} := K_B \cdot \frac{8 \cdot F_{Closed} \cdot D}{\pi \cdot d^3} = 121.691\ MPa$$

$$\tau_m := \frac{\tau_{upper} + \tau_{lower}}{2} = 243.108\ MPa$$ Mean stress

$$\tau_a := \frac{\tau_{upper} - \tau_{lower}}{2} = 121.418\ MPa$$

Average Stress

$$S_u := \tau_{max} = \left(1.106 \cdot 10^3\right)\ MPa$$

$$S_e := 0.4 \cdot S_u = 442.412\ MPa$$

$$n_f := \frac{1}{\dfrac{\tau_a}{S_e} + \dfrac{\tau_m}{S_u}} = 2.023$$

Fatigue Factor of Safety